

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

$f(x) = 4x + b$

For the linear function f , b is a constant and $f(7) = 28$. What is the value of b ?

Answer

- A. 0
- B. 1
- C. 4
- D. 7

Correct Answer: A

Rationale

Choice A is correct. For the linear function f , it's given that $f(7) = 28$. Substituting 7 for x and 28 for $f(x)$ in the given function yields $28 = 4(7) + b$, or $28 = 28 + b$. Subtracting 28 from each side of this equation yields $0 = b$. Therefore, the value of b is 0.

Choice B is incorrect. Substituting 1 for b in the given function yields $f(x) = 4x + 1$. For this function, when the value of x is 7, the value of $f(x)$ is 29, not 28.

Choice C is incorrect. Substituting 4 for b in the given function yields $f(x) = 4x + 4$. For this function, when the value of x is 7, the value of $f(x)$ is 32, not 28.

Choice D is incorrect. Substituting 7 for b in the given function yields $f(x) = 4x + 7$. For this function, when the value of x is 7, the value of $f(x)$ is 35, not 28.